Stage 1 Test Report: Raw Material & Fabric Destructive Test Report

Destructive Material Evaluation of Drop-Stitch Fabric & PVC Coating for Felizbay Inflatable Pool (400 cm × 300 cm × 120 cm, 20 cm Wall Thickness)

1. Product Specifications, Material Traceability & Test Scenario

Report No.	`RPT-FB-QC-20261113-S1`	**Governing SOP**	`WI-QC-DS-2026-001 (V1.0)` Stage 1
Brand / Model	**Felizbay** Large Rectangular Drop-Stitch Pool (`FB-DSP-400`)	**Dimensions / Wall**	**400 cm × 300 cm × 120 cm** / Wall Thickness **20 cm**
Bonding Method	**RF / Hot-Air Heat Welding + 2-Part PU Cold-Bonding Adhesive**	**Rated Pressure**	**P_W = 8.0 PSI** (0.55 bar)
Sample Batch / Unit	Batch `LOT-202611-A08` / **Unit #1** (Material Coupon Cutout)	**Fabric Lot Trace**	1000D High-Density Drop-Stitch + 0.9 mm PVC (`HY-TEX` / Roll `R261029-04B`)
Test Equipment	5 kN Universal Testing Machine, Φ150 mm Puncture Ring, -25°C Freezer	**Condition / Result**	23.2°C, 52% RH / **Stage 1 Approved (PASS)**

Test Scenario & Objective: Conducted inside the climate-controlled physical testing laboratory (23.2°C), base fabric and composite seam coupons were excised directly from **Felizbay Unit #1** (400 × 300 × 120 cm, 20 cm wall thickness). By isolating the raw substrate from full-assembly geometry, Stage 1 verifies PVC airtight coating adhesion, pressurized puncture tear resistance, and -18°C cold-crack resistance.

2. Stage 1 Specimen Preparation & Destructive Testing Photographs



Fig. 1-1: Coupon Cutting & 120°C Hot-Air Pre-Peeling (Test 1.1)
Twenty 250 × 50 mm strips cut ≥ 50 mm from seams, marked for Warp/Weft, and softened at 120°C for 5–8 s to pre-peel a 50 mm gripping tab without nicking base yarns.



Fig. 1-2: Universal Testing Machine 180° Peel Test (Test 1.1)
Pneumatic serrated grips set at 100 mm initial gauge clamp the blue PVC coating and white polyester substrate at 180° alignment, pulling at 100 mm/min over a 150 mm stroke.

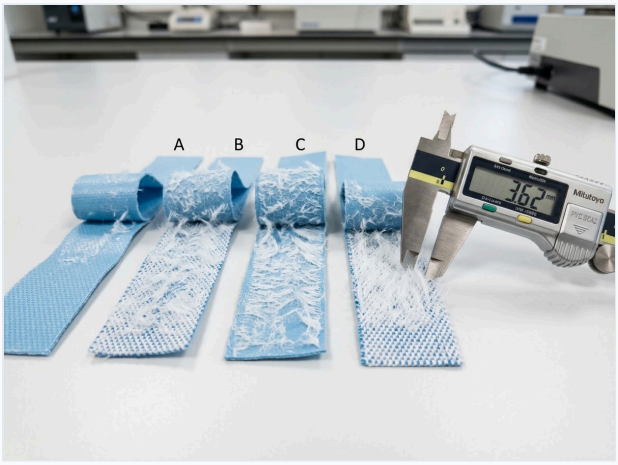


Fig. 1-3: Post-Peel Fiber Cohesive Failure & Caliper Check (Test 1.1)
Peeled PVC coating backing retains dense, torn white polyester substrate fibers with zero smooth interfacial delamination, confirming thorough resin penetration.

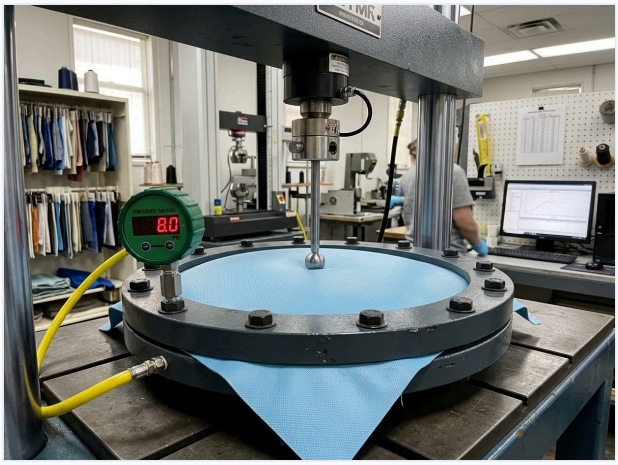


Fig. 1-4: 8.0 PSI Pressurized Single-Point Puncture Test (Test 1.2)
250 × 250 mm floor composite clamped in a Φ150 mm pneumatic ring fixture at 8.0 PSI and penetrated vertically at 50 mm/min by an R1.0 mm hemispherical steel stylus.

3. Detailed Test Parameters & Measured Data Records

3.1 Test 1.1: Airtight Coating 180° Peel & Base Fabric Tear Test (All 20 Specimens)

Test Parameters: 5 Warp and 5 Weft strips each from the 20 cm wall and 10 cm reinforced floor (20 specimens total); 150 mm peel stroke at 100 mm/min. Initial and final 25 mm transient zones are excluded; the middle **100 mm steady-state zone** determines mean peel strength (Spec: ≥ 120 N/5cm) and minimum trough value (Spec: ≥ 100 N/5cm).

Location & Axis	Specimen IDs	Steady-State Mean Peel (N/5cm)	Minimum Trough (N/5cm)	Group Mean	Min Trough	Failure Mode	Result
Wall — Warp (20 cm Wall)	`W-W1 ~ W-W5`	`142.6 / 139.8 / 145.2 / 141.0 / 143.4`	`124.5 / 121.0 / 128.3 / 122.8 / 125.1`	**142.4**	**121.0**	Cohesive Fiber Tear	**PASS**
Wall — Weft (20 cm Wall)	`W-F1 ~ W-F5`	`136.5 / 138.2 / 134.9 / 137.4 / 135.5`	`118.2 / 119.6 / 115.4 / 117.9 / 116.8`	**136.5**	**115.4**	Cohesive Fiber Tear	**PASS**
Floor — Warp (Wear Layer)	`F-W1 ~ F-W5`	`148.5 / 151.2 / 147.8 / 150.1 / 149.4`	`131.0 / 133.5 / 129.8 / 132.2 / 130.6`	**149.4**	**129.8**	Cluster Yarn Rupture	**PASS**
Floor — Weft (Wear Layer)	`F-F1 ~ F-F5`	`141.8 / 140.2 / 143.6 / 139.5 / 141.9`	`123.4 / 121.9 / 125.0 / 120.8 / 122.7`	**141.4**	**120.8**	Cohesive Fiber Tear	**PASS**

3.2 Test 1.2 (Pressurized Puncture) & Test 1.3 (-18°C Low-Temperature Folding) Results

Test Sub-Item & Parameters	Sample ID	Factory Acceptance Criteria	Measured Peak Data & Observation	Result
Test 1.2 Pressurized Puncture (8.0 PSI / R1.0 mm Stylus / 50 mm/min)	`P-01 (Floor)`	Peak puncture force **≥ 350 N**; Total warp/weft tear propagation **≤ 30 mm**, no burst	Hold drop 0.0 PSI; Peak force **418.5 N**; Total tear length **16.4 mm**	**PASS**
	`P-02 (Floor)`		Hold drop 0.1 PSI; Peak force **406.2 N**; Total tear length **18.2 mm**	**PASS**
	`P-03 (Floor)`		Hold drop 0.0 PSI; Peak force **425.0 N**; Total tear length **15.8 mm**	**PASS**
Test 1.3 -18°C Cold Folding (4h Freeze / 5 kg Cross-Fold / 10 PSI Leak)	`LT-B1 ~ B3 (3 Panel Cuts)`	**Zero stress-whitening or cracking** at fold intersection; **0 bubbles** for 3 min at 10.0 PSI	Coating remained pliable without whitening or micro-cracks; 0 bubbles in 3 min	**PASS**
	`LT-S1 ~ S3 (3 Seam Cuts)`		Heat-welded + PU adhesive composite edge tapes showed zero delamination; 0 bubbles	**PASS**

4. Stage 1 Conclusion & Material Release Disposition

Stage 1 Summary Conclusion: Raw material coupons from the **Felizbay** Drop-Stitch Inflatable Pool (`FB-DSP-400`, Batch `LOT-202611-A08`, Fabric Roll `R261029-04B`) passed all Stage 1 destructive tests with substantial safety margins—achieving 136.5–149.4 N/5cm mean peel strength (100% cohesive fiber tear), 406.2–425.0 N pressurized puncture resistance, and zero cold-cracking at -18°C. The 1000D drop-stitch fabric lot is free of raw material defects and approved for production release.

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